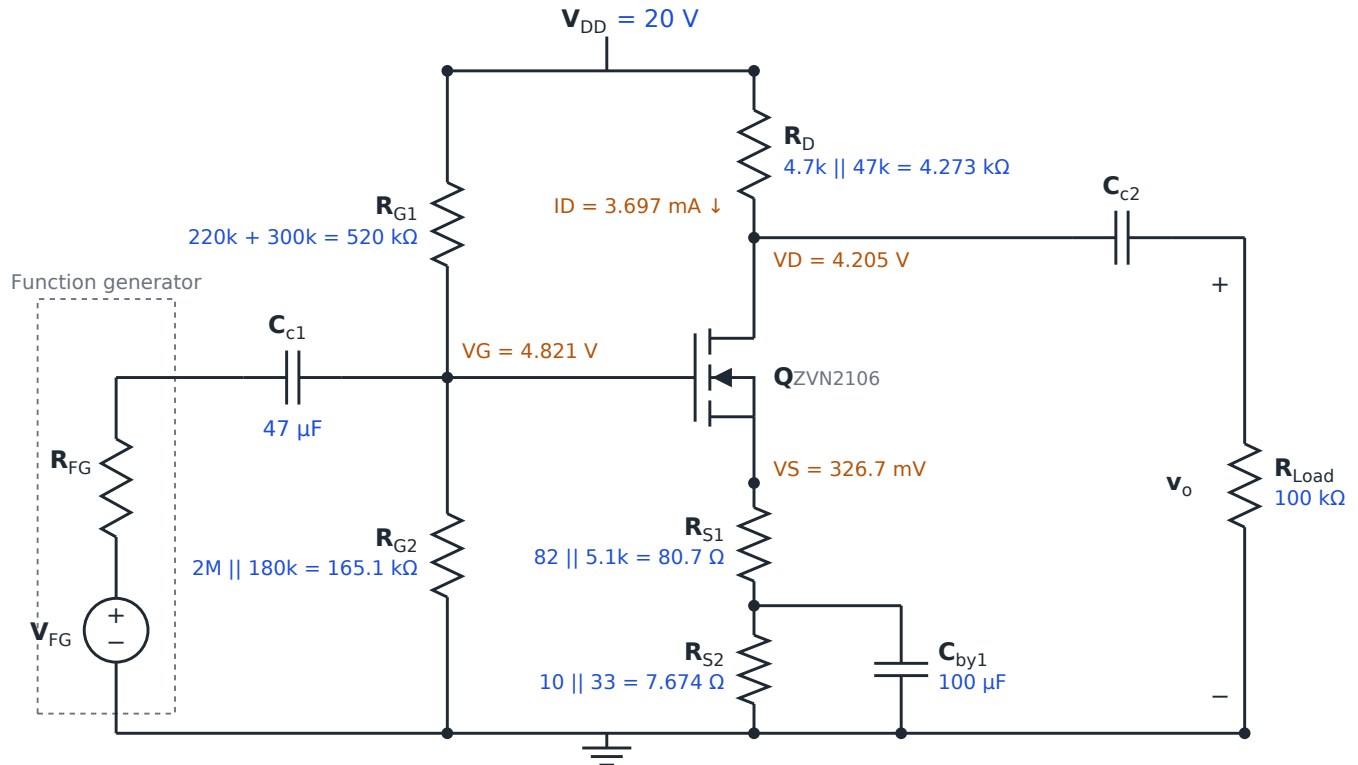


Common-Source Stage — Optimized Design

Nominal score 0.9316

ECE321 Lab 3 · generated 2026-10-08 07:25



■ component values (from Circuit_Sol)

■ DC operating point (from analyzeCSAmpModel)

Cc2 value not part of model

Analyzed values

Quantity	Nominal
Overdrive voltage V_{OV}	762.9 mV
Drain current I_D	3.697 mA
Gate voltage V_G	4.821 V
Source voltage V_S	326.7 mV
Drain voltage V_D	4.205 V
Drain-source V_{DS}	3.879 V
Gate-source V_{GS}	4.494 V
Transconductance g_m	9.69 mS
Input resistance R_i	125.3 kΩ
Open-circuit gain A_{v_o}	-23.23 V/V
Output resistance R_o	4.273 kΩ
Loaded gain A_v	-22.27 V/V
Power R_D	53.08 mW
Power R_{G1}	255.6 μW
Power R_{G2}	129.1 μW
Power R_{S1}	1.085 mW
Power R_{S2}	80.48 μW
Swing margin, downward	1.516 V
Swing margin, upward	13.55 V
Distortion estimate HD_2	1.32 %
Threshold voltage V_t	2 V

Score breakdown (nominal)

Category	Score	Weight
ID in range	0.9522	1
RI in range	0.9987	1
RO in range	0.9162	1
Av in range	0.9970	1
In saturation	1.0000	1
Saturation margin	1.0000	1
RG1 Power	0.2500	0.25
RG2 Power	0.2500	0.25
RS1 Power	0.2499	0.25
RS2 Power	0.2500	0.25
RD Power	0.2429	0.25
Low Distortion	0.5791	1
Meets req	1.0000	1
Total (nominal)	0.9316	